SilverSteps: tech stack and method

How the prototype is built, and how it finds the footpaths where seniors face barriers.

Tech stack

Layer	What we use
Data pipeline	Python 3.11 with **osmnx** (OpenStreetMap walking network), **networkx** (routing), **geopandas** and **shapely** (spatial joins and distances), **pandas**, scikit-learn (nearest-node lookups)
AI photo checks	**Claude Opus 5.5** via the Anthropic Python SDK, with structured JSON output, reading **Google Street View Static API** images
Website	**React 19 + TypeScript**, built with **Vite**. **MapLibre GL** draws the map over **OneMap** basemap tiles. Phosphor icons, Public Sans font, plain CSS with light and dark themes
Backend	**FastAPI** (Python) serving the site plus an API for corrections and resident routes. **SQLAlchemy** with **MariaDB** in production, SQLite locally
Hosting	**Deployer** (self-hosted, Docker) at silversteps.liewjiaen.com. **Firebase Hosting** static backup. A **Cloudflare Worker** switches to the backup automatically if the main server is down

The model is precomputed into static JSON per region, so the site needs no server-side computation. The server only stores feedback.

How we find the places with issues

1. Estimate where seniors live

- Each subzone's residents aged 65+ (SingStat, June 2025) are scaled by the share living in HDB flats.
- That total is then spread across the subzone's HDB blocks by flat count.

2. Estimate where they walk (the footfall)

- Every footpath segment gets a senior walking cost, which is its length adjusted for difficulty:
  - steps × 4
  - sheltered × 0.8
  - walking on a road with no separate footpath × 1.25
  - +40 m for a crossing with no signals, +20 m for a signalised one
- From every block, find all reachable places within a cost of 800 m.
- For each type of place (hawker, bus stop, clinic and so on), seniors pick one. Nearer ones are chosen more often (weight = $e^{-\text{cost} / 300\ \text{m}}$).
- Trips per senior per week are assumptions: hawker 5, bus stop 4, kopitiam 3, park and supermarket 2, MRT and eldercare 1.5, community club and fitness corner 1, clinic 0.5, polyclinic 0.25. Each trip counts both ways.
- Adding every trip onto the paths it uses gives **estimated senior trips per week on every segment**.

3. Check each busy path for barriers

Barrier	How it's detected	Rule it's checked against
Steps	OSM `highway=steps`	BCA: accessible routes exclude steps
Slope / too steep	OSM `incline` gradient; steeper than 1:12 is severe	BCA Tables 4 and 5
Raised kerb	OSM `kerb=raised`	BCA: flush or kerb ramp, at most 1:10
Crossing without signals	OSM crossing type	LTA
No tactile paving / silent traffic light	OSM `tactile_paving`, `traffic_signals:sound`	BCA cl. 4.5.3.3 and 4.7.4.1
Rough surface, unlit, obstruction	OSM tags plus Street View photos	BCA cl. 4.1.1, 9.1 and 4.2.3
No seat	No bench or bus stop within 25 m of the path's midpoint (gap over 50 m); worse past 100 m (over 200 m)	BCA: resting areas at most 50 m apart
No shelter	OSM `covered`, plus Street View	BCA, and Green Mark sheltered routes to transport

- **Comfort gaps** (no seat, no shelter) are flagged only on the busiest quarter of paths, because OSM records few benches. The exception is paths within 150 m of a bus stop or MRT, which are always checked for shelter.
- **Unrecorded attributes** (shelter, lighting, tactile paving, signal audio) show as "needs a field check", not as barriers.
- **Photo checks:** Claude reviews two Street View photos of each region's 30 busiest paths. Only findings with confidence of at least 0.6 add or remove barriers.

4. Turn barriers into ranked fixes

- Connected path pieces with the same barrier merge into one fix, labelled by the nearest block, e.g. "near Blk 475 Ang Mo Kio Ave 10".
- **Rank = estimated senior trips per week × severity.** Severity is 3 for steps or a too-steep slope, 2 for crossings, kerbs, tactile paving, obstructions and long seat gaps, and 1 for comfort issues.
- Works already planned in HDB Town Design Guides are tagged and moved down the list.
- **High-risk areas (A to E):** the worst fix plus every fix within 150 m, scored by total trips × severity.

5. Residents correct it live

- "Wrong" removes a barrier, "Missed" adds one, and the ranking recomputes instantly.
- Drawn routes are compared with the model to give an accuracy figure.